

NYU

Using the PATH study to describe the trajectory of, and relationship between, e-cigarette use and internalizing mental health disorders

GPH Tobacco Research Lab

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Introduction

EC and IMHD

- While cigarette use in the US continues to decline, there has been a remarkably rapid uptake of electronic cigarette use (EC).
- Previous studies suggested that EC is associated with internalizing mental health disorders (IMHD) which can dramatically impair quality of life and overall health. However, **few studies dive into the causal effect of EC on Internalizing mental health disorders, and none of them discuss the bidirectional relationship.**
- According to a recent systematic review study (Becker et al.2021), well-designed longitudinal studies to investigate long-term mental health sequelae of EC remain needed since many existing studies are cross-sectional, directionality remains uncertain.

PATH study

- The PATH study is designed to collect information about the use of tobacco products, tobacco dependence, cessation, perceptions of risk and harm, overall physical and mental health, peer and family influences, other substance use, and demographic information (Hyland et al., 2017; ICPSR, 2019), which is a large, nationally representative, cohort study.
- Previous studies suggested high-severity past-year internalizing problems (compared to low severity) significantly related to current EC use (aOR=1.97, 95% CI: 1.46-2.65, $p<0.001$) in cross-sectional analysis of baseline PATH data (Ganz et.al 2020), and high severity lifetime internalizing problems predicted onset of EC (aOR=1.4, 95% CI: 1.1-1.8, $p<0.05$) among non-users in a one-year longitudinal analysis (Green et.al 2018). **None of them describe the trajectory of, and relationship between EC and IMHD using multiple waves.**

Study purpose

Aim 1: To describe the average trajectory of EC and IMHD among US adults from 2011 to 2021

Aim 2: To explore the confounding effect of several socio-demographic characteristics (e.g. age, race, poverty, education) and substance use (e.g. alcohol, illicit drugs, or other combustible tobacco, marijuana)

Aim 3: To assess the longitudinal bidirectional relationship between EC and IMHD utilizing the cross-lagged model.

Methods

Study participants

This study uses data from N=13,660 adults participating in wave 1-6 (2011-2021) of the PATH study, who have at least 2-wave completed records of EC and internalizing mental health assessment (GAIN-SS) across 6 waves.

Study design

Longitudinal study utilizing 6-wave adult PATH data, where

2013 -- 2014 (Wave 1), 2014 -- 2015 (Wave 2), 2015 -- 2016 (Wave 3), 2016 -- 2018 (Wave 4), 2018 -- 2019 (Wave 5), 2021 (Wave 6)

Measures

- Time-invariant: baseline age group, gender, race/ethnicity, poverty (>50,000), education level
- Time-varying: current EC, IMHD (GAIN>=2), alcohol use (30 days), marijuana use (30 days), drug use (12 months), other current tobacco use (other combustible use)
- Missing value: if <10% missing, NA; if >10%, keep as a separate category

Methods

Statistical analysis (R, Mplus)

Aim 1: To describe the average trajectory of EC and IMHD among US adults from 2011 to 2021

- Sample description table（baseline)
- Prevalence of EC and IMHD in each wave and unconditional population average logistic regression
- Nonparametric individual trajectories for EC and IMHD

Aim 2: To explore the confounding effect of several socio-demographic characteristics and substance use

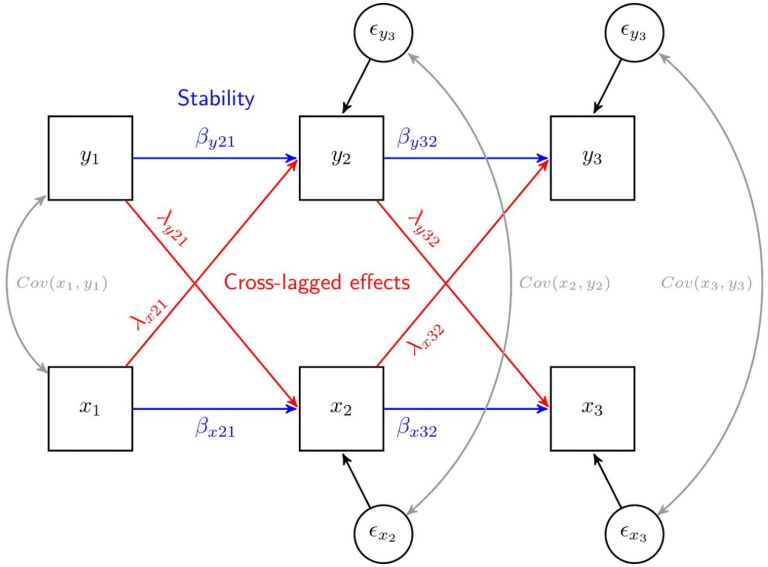
- Bivariate tables (baseline) with chi-square p-value
- Random intercept logistic regression model

Aim 3: To assess the longitudinal bidirectional relationship between EC and IMHD utilizing the cross-lagged model.

- Cross-lagged model

$$X_{t+1} = \lambda_{y_{t+1,t}} * y_t + \beta_{x_{t+1,t}} * x_t + \epsilon_{x_{t+1}}$$

$$Y_{t+1} = \lambda_{x_{t+1,t}} * x_t + \beta_{y_{t+1,t}} * y_t + \epsilon_{y_{t+1}}$$



β: auto-relationship

λ: cross-lagged relationship

cov(x,y): covariance

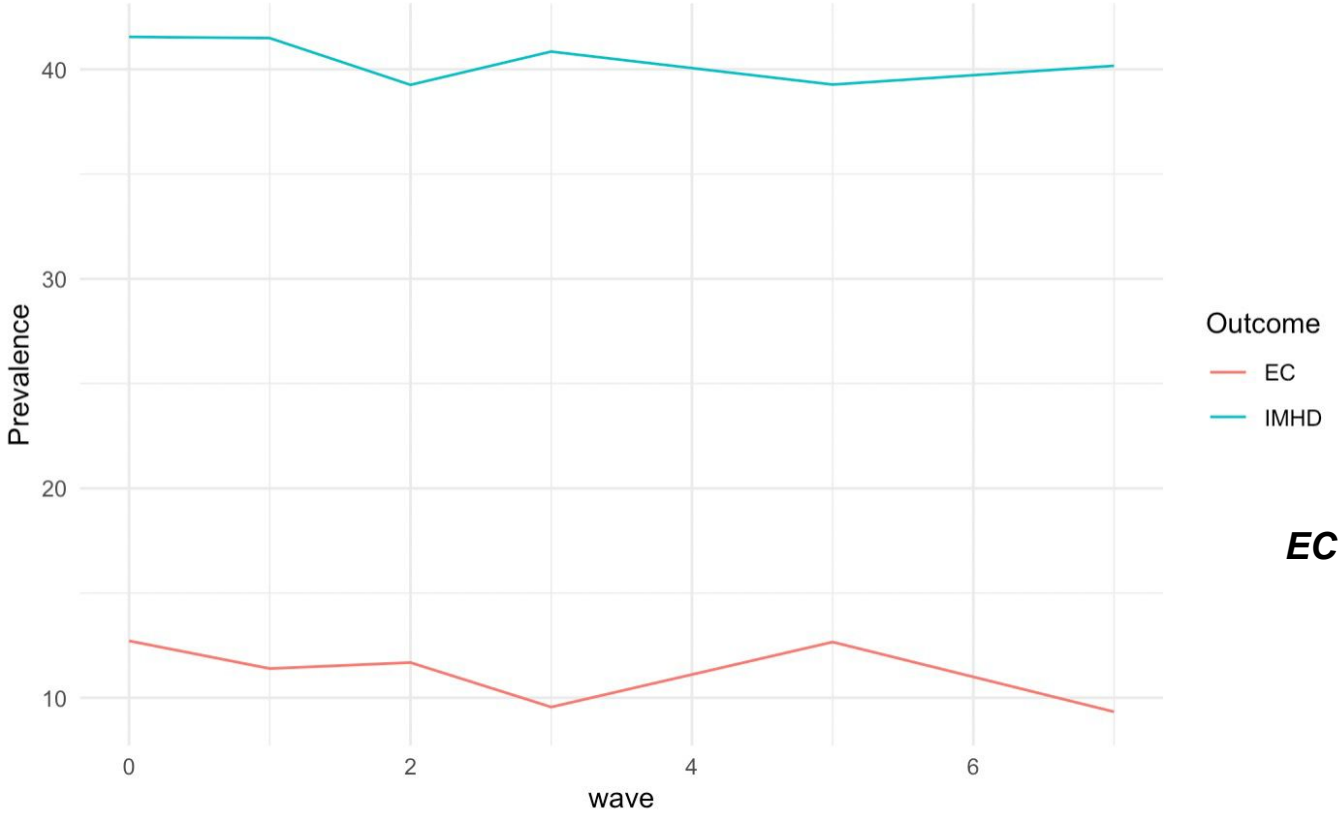
ε: error

Results: Baseline sample description table (N=13,660)

	N (%)
Age group	
18-24	3420 (25.0%)
25-34	2775 (20.3%)
35-44	2237 (16.4%)
45-54	2329 (17.0%)
55-64	1862 (13.6%)
>65	1037 (7.6%)
Gender	
Female	7360 (53.9%)
Male	6300 (46.1%)
Race/ethnicity	
Hispanic	2331 (17.1%)
Non-hispanic white	8237 (60.3%)
Non-hispanic black	2117 (15.5%)
Others	975 (7.1%)
Poverty	
No	5003 (36.6%)
Yes	8657 (63.4%)
Education level	
Less than high school	1553 (11.4%)
GED	832 (6.1%)
High school graduate	2864 (21.0%)
Some college (no degree) or associate's degree	4901 (35.9%)
Bachelor degree or advanced degree	3510 (25.7%)

	N (%)
EC Ever	
No	9031 (66.1%)
Yes	4629 (33.9%)
Alcohol	
No	1913 (14.0%)
Yes	11747 (86.0%)
Mariju	
No	6404 (46.9%)
Yes	3819 (28.0%)
Unknown	3437 (25.2%)
Drug use	
No	9313 (68.2%)
Yes	4347 (31.8%)
Moderate/severe IMHD	
No	7984 (58.4%)
Yes	5676 (41.6%)
Combustible tobacco use	
Never	5023 (36.8%)
Former	6301 (46.1%)
Current	2336 (17.1%)

Results: Prevalence and population average model of EC and IMHD




```
Deviance Residuals:
   Min      1Q  Median      3Q     Max
-1.029  -1.022  -1.003   1.342   1.362

Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept) -0.359347   0.011437 -31.418  < 2e-16 ***
wave        -0.009381   0.002995  -3.133  0.00173 **
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 110494  on 81879  degrees of freedom
Residual deviance: 110484  on 81878  degrees of freedom
  (80 observations deleted due to missingness)
AIC: 110488
```

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Deviance Residuals:
    Min       1Q   Median       3Q      Max
-0.5085  -0.5015  -0.4876  -0.4608   2.1426

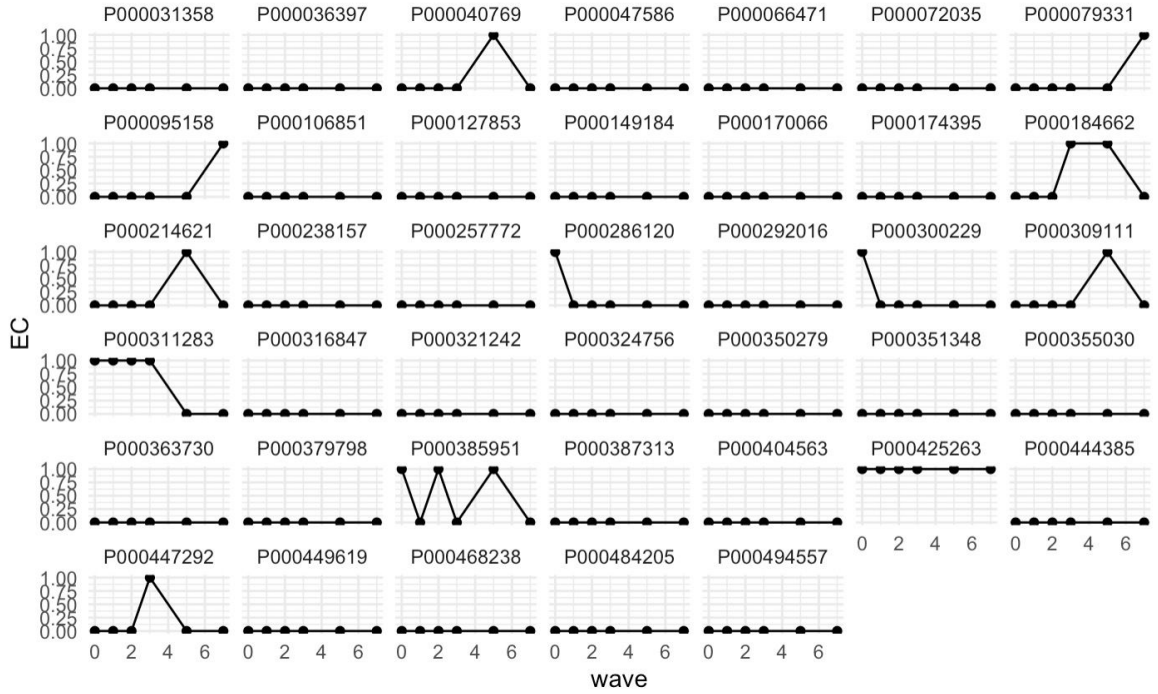
Coefficients:
             Estimate Std. Error  z value Pr(>|z|)
(Intercept) -1.98016    0.01754 -112.875  < 2e-16 ***
wave        -0.02987    0.00472   -6.329 2.47e-10 ***
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Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

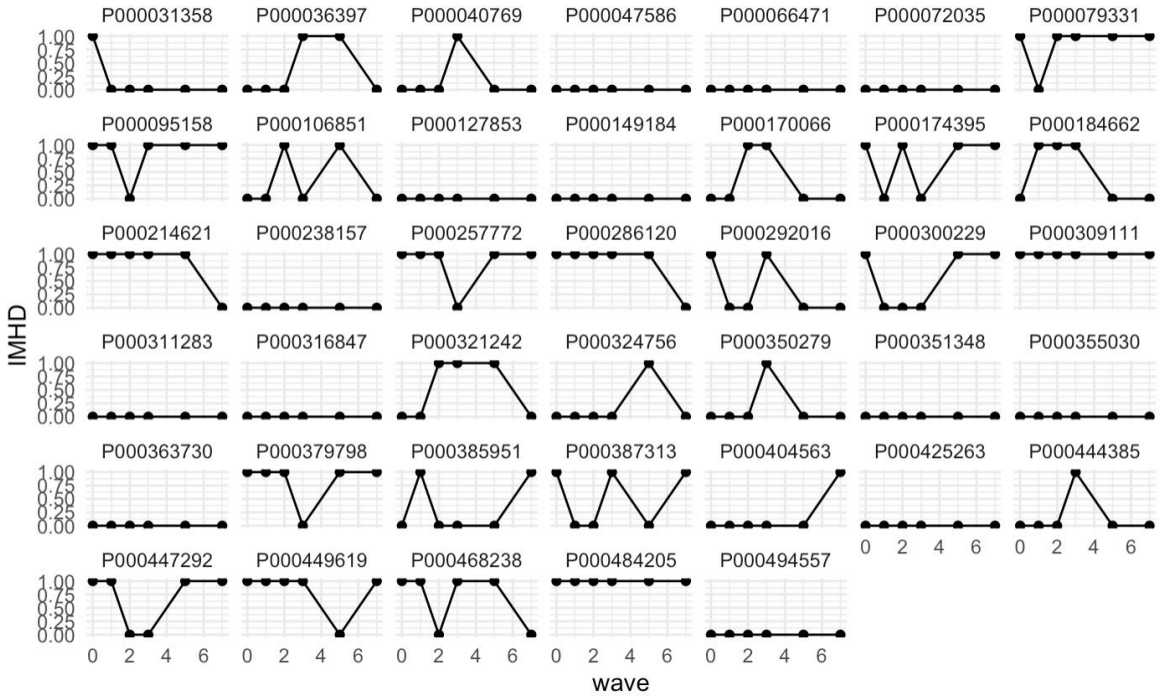
    Null deviance: 57276  on 81550  degrees of freedom
Residual deviance: 57235  on 81549  degrees of freedom
  (409 observations deleted due to missingness)
AIC: 57239
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Results: Individual trajectories for EC and IMHD

EC current use



IMHD



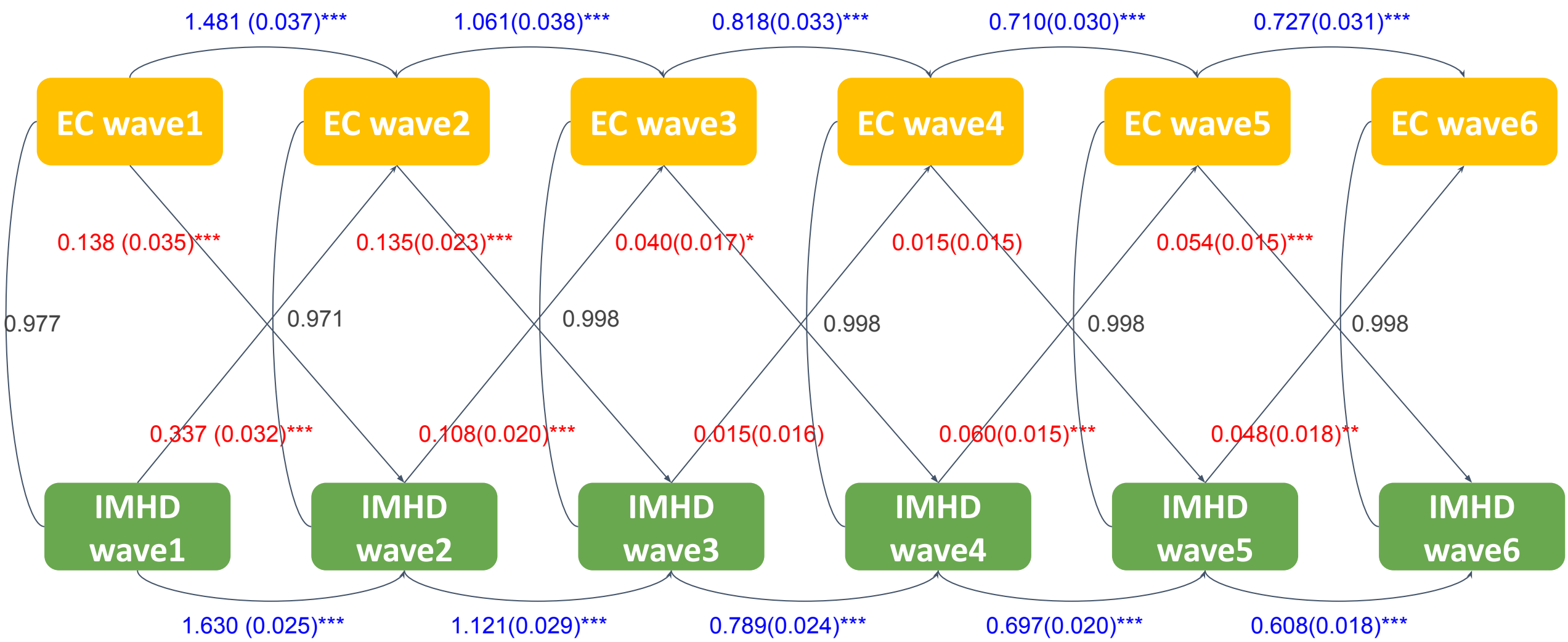
Results: Random intercept logistic Regression

		Unconditional model	Conditional model with time-invariant covariates	Conditional model with time-invariant covariates and time-varying covariates
Fixed effects				
Initial status, π_{0i}	intercept	-0.720 (0.00007)***	-1.308 (0.102)***	-1.089 (0.099)***
	EC (Yes)	0.547 (0.00007)***	0.392 (0.054)***	0.345 (0.060)***
	Age			
	25-34		-0.310 (0.059)***	-0.217 (0.060)***
	35-44		-0.379 (0.064)***	-0.258 (0.064)***
	45-54		-0.522 (0.065)***	-0.419 (0.064)***
	55-64		-0.844 (0.070)***	-0.840 (0.071)***
	>64		-1.564 (0.090)***	-1.546 (0.091)***
	Gender(male)		-0.903 (0.042)***	-0.879 (0.042)***
	Race			
	White		0.225 (0.057)***	0.454 (0.057)***
	Black		-0.088 (0.070)	-0.062 (0.071)
	Others		0.409 (0.089)***	0.569 (0.090)***
	Poverty (Yes)		0.584 (0.047)***	0.647 (0.048)***
	Education			
	GED		0.039 (0.099)	0.040 (0.100)
	High school		-0.099 (0.073)	-0.229 (0.074)**
	Some college		0.019 (0.069)	0.017 (0.070)
	Bachelor+		-0.410 (0.078)***	-0.453 (0.078)***
	Alcohol (Yes)		0.526 (0.063)***	0.321 (0.045)***
	Marijuana (Yes)		0.465 (0.053)***	0.275 (0.045)***
	Drug use (Yes)		0.709 (0.047)***	0.599 (0.050)***
	Other tobacco use			0.239 (0.032)***
	Former		0.145 (0.051)**	
	Current		0.260 (0.067)***	

Results: Random intercept logistic Regression

		Unconditional model	Conditional model with time-invariant covariates	Conditional model with time-invariant covariates and time-varying covariates
Rate of change, π_{1i}	intercept	-0.009 (0.00007)***	-0.009 (0.005)	0.012 (0.008)
	EC	-0.0447 (0.00007)***	-0.064 (0.016)***	-0.052 (0.015)***
	Alcohol			0.013 (0.009)
	Marijuana			0.053 (0.011)***
	Drug use			0.043 (0.015)**
	Other tobacco use			-0.016 (0.007)*
Variance Components				
Level 1	within-person			
Level 2	Initial status	4.638 (2.154)	3.69 (1.921)	3.764 (1.940)
	Rate of change			
	Deviance	89955.7	87602.4	83664.6
	AIC	89965.7	87652.4	83718.6
	BIC	90012.3	87885.1	83968.7

Results: Cross-lagged model



Summary

Preliminary findings

- On average, the odds of e-cigarette smoke at baseline is 0.15, indicating a lower probability of e-cigarette use at the baseline of wave. There's evidence that on average the probability of e-cigarette use declines across successive waves.
- On average, the odds of moderate or severe internalizing mental health disorder at baseline is 0.7, indicating a lower probability of IMHD at the baseline of wave. There's evidence that on average the probability of IMHD declines across successive waves.

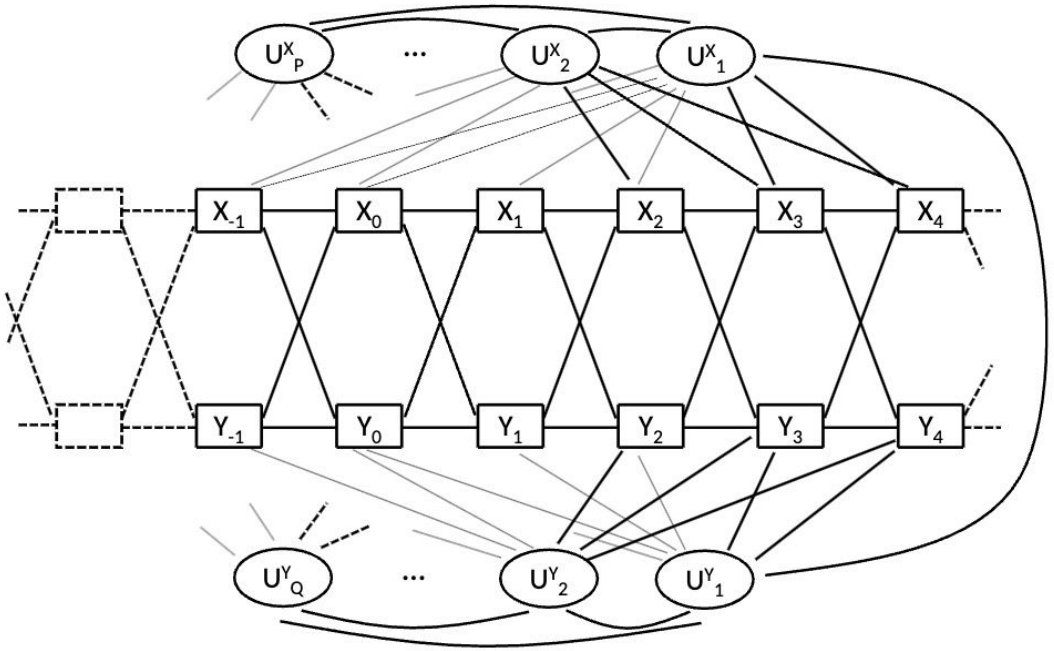
- Being female with lower age (younger), lower education level, poverty, substance use related to higher odds of long-term IMHD.
- Drug and marijauna use increase the odds of IMHD over time.

- E-cigarette use will increase the odds to have worse internalizing mental health status.
- Internalizing mental health disorders will increase the odds to use E-cigarette.

Next steps

- Continuous GAIN to measure IMHD without specifying threshold
- Adjusted cross-lagged model
  - U^X_P=time-invariant covariate P affecting X, U^y_Q=time-invariant covariate Q affecting Y
  - $$X_{t+1} = \lambda_{y_{t+1,t}} * y_t + \beta_{x_{t+1,t}} * x_t + f_U{}^x(U^x{}_1,\ldots, U^x{}_p) + \epsilon_{x_{t+1}}$$

    $$Y_{t+1} = \lambda_{x_{t+1,t}} * y_t + \beta_{x_{t+1,t}} * y_t + f_U{}^y(U^y{}_1,\ldots, U^y{}_p) + \epsilon_{x_{t+1}}$$
  - Also time-varying covariates…
- Subgroup analysis
  - baseline IMHD severe/moderate vs mild/none
  - History of other tobacco product usage Yes vs No
  - Covid-19 outbreak before vs after



Thank you!

